

copy three-dimensional figures, assemble blocks, or arrange sticks in specific designs)

(4) personality change, i.e., alteration or accentuation of premorbid traits

D. State of consciousness not clouded (i.e., does not meet the criteria for Delirium or Intoxication, although these may be superimposed).

E. Either (1) or (2):

(1) evidence from the history, physical examination, or laboratory tests, of a specific organic factor that is judged to be etiologically related to the disturbance

(2) in the absence of such evidence, an organic factor necessary for the development of the syndrome can be presumed if conditions other than Organic Mental Disorders have been reasonably excluded and if the behavioral change represents cognitive impairment in a variety of areas

Amnestic Syndrome

The essential feature is impairment in short- and long-term memory occurring in a normal state of consciousness (i.e., not clouded). The disturbance is attributed to a specific organic factor. Amnestic Syndrome is not diagnosed if memory impairment exists in the context of clouded consciousness (Delirium) or in association with a more general loss of intellectual abilities (Dementia).

The individual with an Amnestic Syndrome has both an ongoing inability to learn new material (short-term memory deficit; anterograde amnesia) and an inability to recall material that was known in the past (long-term memory deficit; retrograde amnesia). The former is conventionally assessed by requiring the individual to remember several unrelated words or a short paragraph after a brief (usually 5-15-minute) interval of distraction. The latter is tested by asking questions about events of the past such as birthplace, family, schooling, vocation, major historical events, the names of recent presidents, etc. The individual with an Amnestic Syndrome has difficulty with both of these operations of memory. Events of the very remote past, however, are often better recalled than more recent events. For example, an individual may remember in vivid detail a hospital stay that took place a decade before examination, but may have no idea that he or she is currently in the hospital. So-called "immediate memory" (e.g., digit span), however, is *not* impaired in Amnestic Syndrome.

Associated features. A significant degree of amnesia nearly always results in disorientation. Confabulation, the recitation of imaginary events to fill in gaps in memory, is often observed, and when present tends to disappear with time. Most individuals with this syndrome lack insight into their memory deficit, and may explicitly deny it, despite evidence to the contrary. Others acknowledge a problem, but appear unconcerned. Apathy, lack of initiative, and emotional

blandness are common. Although the individual is superficially friendly and agreeable, his or her affect is shallow.

When Amnestic Syndrome is the result of Alcohol Dependence and vitamin deficiency (see Alcohol Amnestic Disorder, p. 136), other neurological complications of alcohol ingestion and malnutrition, such as peripheral neuropathy, cerebellar ataxia, etc., may also be observed.

Course. The mode of onset depends on the etiology. In most cases it is fairly sudden. The subsequent course, also a function of the etiology, is usually one of chronicity.

Impairment. Impairment in social and occupational functioning is usually moderate to severe.

Complications. Any complications are the direct result of the individual's memory impairment. For example, the individual's forgetting to extinguish a lighted cigarette may cause a fire.

Etiological factors. Amnestic Syndrome may result from any pathological process that causes bilateral damage to certain diencephalic and medial temporal structures (e.g., mammillary bodies, fornix, hippocampal complex). Examples include head trauma, surgical intervention, hypoxia, infarction in the territory of the posterior cerebral arteries, and herpes simplex encephalitis. The most common form of Amnestic Syndrome is that associated with thiamine deficiency and chronic use of alcohol.

Prevalence. The syndrome is apparently uncommon.

Differential diagnosis. Delirium and Dementia also involve memory impairment. In Delirium, however, there is also a clouding of consciousness; and in Dementia, there are other major intellectual deficits as well.

In **Factitious Disorder with Psychological Symptoms**, memory testing often yields inconsistent results. Furthermore, there is no organic etiologic factor.

Diagnostic criteria for Amnestic Syndrome

A. Both short-term memory impairment (inability to learn new information) and long-term memory impairment (inability to remember information that was known in the past) are the predominant clinical features.

B. No clouding of consciousness, as in Delirium and Intoxication, or general loss of major intellectual abilities, as in Dementia.

C. Evidence, from the history, physical examination, or laboratory tests, of a specific organic factor that is judged to be etiologically related to the disturbance.

Organic Delusional Syndrome

The essential feature is the presence of delusions that occur in a normal state of consciousness and that are due to a specific organic factor. The diagnosis is not made if delusions occur in a clouded state of consciousness, as in Delirium, if there is a significant loss of intellectual abilities, as in Dementia, or prominent hallucinations are present, as in Organic Hallucinoses.

The nature of the delusions is variable and depends, to some extent, on the etiology. Persecutory delusions are the most common type. Amphetamine use may cause a highly organized paranoid delusional state indistinguishable from the active phase of Schizophrenia. Some individuals with cerebral lesions develop the delusion that a limb of their body is missing.

Hallucinations may be present in an Organic Delusional Syndrome, but they are not the predominant feature.

Associated features. Mild cognitive impairment is often observed. As in Schizophrenia, almost any symptom may occur as an associated feature. The individual may appear perplexed, disheveled, or eccentrically dressed. Speech may be rambling or incoherent. Abnormalities of psychomotor activity may occur, with either hyperactivity (pacing, rocking), or apathetic immobility. Ritualistic, stereotyped behavior, sometimes associated with magical thinking, may also be observed. A dysphoric mood is common.

Impairment. Impairment in social and occupational functioning is usually severe.

Complications. The individual may harm himself or herself or others while reacting to delusions.

Etiological factors. These are diverse. A number of substances—e.g., amphetamines, cannabis, and hallucinogens—may cause this syndrome. Some individuals with temporal lobe epilepsy have an interictal Organic Delusional Syndrome that is almost indistinguishable from Schizophrenia. A paranoid Organic Delusional Syndrome has been described in some cases of Huntington's chorea. Certain cerebral lesions, particularly of the nondominant hemisphere, result in this syndrome.

Certain substances can also cause a more or less permanent Organic Delusional Syndrome, essentially identical with Schizophrenia, even after the substance is no longer present in the body, analogous to "flashback" hallucinations.

Differential diagnosis. Nonorganic psychotic disorders such as Schizophrenia or Paranoid Disorders must be distinguished from Organic Delusional Syndrome. Differentiation rests primarily on evidence, gathered from the history, physical examination, or laboratory tests, of a specific organic factor judged to be responsible for the development of the delusions. The appearance of delusions *de novo* in an individual over the age of 30 years without a known history of Schizophrenia or Paranoid Disorder should always alert the diagnostician to the possibility of an Organic Delusional Syndrome. On the other hand, the fact

that an individual has a prior history of nonorganic psychosis does not mean that one should neglect consideration of an Organic Delusional Syndrome, especially if there is concern about a possible organic factor (for example, the ingestion of an hallucinogen).

In **Organic Hallucinosi**s, hallucinations are the predominant feature. Delusions, if present, are related to the hallucinations. In **Organic Affective Syndrome**, symptoms resembling those of the Affective Disorders predominate. Delusions and hallucinations, if present, have a content related to the mood disturbance.

Diagnostic criteria for Organic Delusional Syndrome

A. Delusions are the predominant clinical feature.

B. There is no clouding of consciousness, as in Delirium; there is no significant loss of intellectual abilities, as in Dementia; there are no prominent hallucinations, as in Organic Hallucinosis.

C. There is evidence, from the history, physical examination, or laboratory tests, of a specific organic factor that is judged to be etiologically related to the disturbance.

Organic Hallucinosis

The essential feature is the presence of persistent or recurrent hallucinations that occur in a normal state of consciousness and that are attributable to a specific organic factor. Therefore, the diagnosis is not made if hallucinations occur in a clouded state of consciousness, as in Delirium, with significant loss of intellectual abilities, as in Dementia, if there is a major disturbance of mood, as in Organic Affective Syndrome, or if delusions predominate, as in Organic Delusional Syndrome.

Hallucinations may occur in any modality, but certain organic factors tend to produce hallucinations of a particular type. For example, hallucinogens most commonly cause visual hallucinations, whereas alcohol tends to induce auditory hallucinations. Individuals who are blind as a result of cataracts may develop visual hallucinations; those who are deaf as a result of otosclerosis will have auditory hallucinations. Hallucinations vary from very simple and unformed to highly complex and organized.

The individual may be aware that the hallucinations are not real, or may have a firm delusional conviction of their reality. Delusions, however, are not the major feature of this syndrome, and are restricted to the content of the hallucinations or to the belief that the hallucinations are real. Further elaboration of delusional material (for example, the development of systematized persecutory delusions to account for the hallucinations, or delusions not related to the hallucinations) suggests an Organic Delusional Syndrome.

Associated features. These features vary according to the etiology, the en-

vironment, and individual differences in response. Hallucinoses in congenial surroundings may be a pleasant experience; in other circumstances, the hallucinations may be fraught with anxiety, depression, or other dysphoric affects.

Course. Course depends on the underlying etiology. Alcohol Hallucinoses may last for as brief a period as a few hours to as long as several years. Untreated cataracts or otosclerosis may cause a chronic Hallucinoses. If the syndrome is the result of sensory deprivation, the duration may be quite brief.

Impairment. The degree of impairment is primarily a function of the underlying etiology.

Complications. Accidents may occur in attempting to flee from frightening hallucinations.

Etiological factors. Use of hallucinogens and prolonged use of alcohol are the most common causes of this syndrome. Sensory deprivation, as in blindness or deafness, is another cause. Seizure foci, especially in the temporal and occipital lobes, may also cause the syndrome.

Differential diagnosis. In **Delirium**, hallucinations, if present, occur with clouding of consciousness. In **Dementia**, hallucinations, if present, are associated with a general loss of intellectual abilities. In **Organic Delusional Syndrome**, hallucinations, if present, are overshadowed by the prominent delusions.

Schizophrenia and **Affective Disorders** may involve hallucinations, but no specific organic factor can be demonstrated.

Hypnagogic and **hypnopompic hallucinations** may occur in individuals without a mental disorder, but they occur only on falling asleep or on awakening.

Diagnostic criteria for Organic Hallucinoses

A. Persistent or recurrent hallucinations are the predominant clinical feature.

B. No clouding of consciousness, as in Delirium; no significant loss of intellectual abilities, as in Dementia; no predominant disturbance of mood, as in Organic Affective Syndrome; no predominant delusions, as in Organic Delusional Syndrome.

C. Evidence, from the history, physical examination, or laboratory tests, of a specific organic factor that is judged to be etiologically related to the disturbance.

Organic Affective Syndrome

The essential feature is a disturbance in mood, resembling either a manic episode or a major depressive episode, that is due to a specific organic factor. The diagnosis is not made if the disturbance in mood occurs in a clouded state of consciousness, as in Delirium; if it is accompanied by a significant loss of intellectual abilities, as in Dementia, or persistent or recurrent hallucinations, as in Organic Hallucinosiis; or if delusions predominate, as in Organic Delusional Syndrome.

The clinical phenomenology of this syndrome is the same as that of a manic or major depressive episode (p. 206 and p. 210). The severity of the disturbance may range from mild to severe. If delusions or hallucinations are present, they are similar to those described under Affective Disorders.

Associated features. Mild cognitive impairment is often observed. Any of the features associated with the Affective Disorders may also be present. If the mood is depressed, these features may include fearfulness, anxiety, irritability, brooding, excessive somatic concerns, phobias, panic attacks, suspiciousness, and a tearful, sad appearance. Delusions of persecution or worthlessness can occur. If the mood is manic, irritability and lability of mood may be present. Hallucinations and delusions are more common in the manic form than in the depressed form.

Impairment. Minimal to severe impairment can result.

Complications. See manic and major depressive episodes (p. 216).

Etiological factors. This syndrome is usually caused by toxic or metabolic factors. Certain substances, notably reserpine, methyl dopa, and some of the hallucinogens, are apt to cause a depressive syndrome. Endocrine disorders are another important etiological factor, and may produce either depressive or manic syndromes. Examples are hyper- and hypothyroidism and hyper- and hypoadrenocorticalism. Carcinoma of the pancreas is sometimes associated with the depressive syndrome, possibly due to an endocrine disturbance. Viral illness may also cause a depressive syndrome. Structural disease of the brain is a rare cause of an Organic Affective Syndrome.

Differential diagnosis. In **Affective Disorders**, no specific organic factor can be demonstrated. When an affective episode follows the taking of a psychoactive substance, such as reserpine, the causal relationship between the ingestion of the substance and the affective disturbance may not be clear. A history of previous Affective Disorder in the individual or in family members suggests that the substance merely triggered an Affective Disorder in an individual who was particularly vulnerable to the organic factor. On the other hand, the absence of a history of previous Affective Disorder in the individual or family members suggests an Organic Affective Syndrome.

In **Organic Personality Syndrome**, there may be a disturbance of mood, but it is not as prominent as the change in personality.

Diagnostic criteria for Organic Affective Syndrome

A. The predominant disturbance is a disturbance in mood, with at least two of the associated symptoms listed in criterion B for manic or major depressive episode (see p. 208 and p. 213).

B. No clouding of consciousness, as in Delirium; no significant loss of intellectual abilities, as in Dementia; no predominant delusions or hallucinations, as in Organic Delusional Syndrome or Organic Hallucinosi.

C. Evidence, from the history, physical examination, or laboratory tests, of a specific organic factor that is judged to be etiologically related to the disturbance.

Organic Personality Syndrome

The essential feature is a marked change in personality that is due to a specific organic factor but that is not due to any other Organic Brain Syndrome. Organic Personality Syndrome in a young child may occur before the development of an enduring style of relating to the environment (personality). In such cases, the syndrome is recognized by significant changes in the child's usual behavior patterns.

The clinical syndrome in a given individual depends principally on the nature and localization of the pathological process. A common pattern is characterized by emotional lability and impairment in impulse control or social judgment. The individual may be belligerent or have temper outbursts and sudden bouts of crying with little or no provocation. Socially inappropriate actions, such as sexual indiscretions, may be engaged in with little concern for the consequences. Another pattern is characterized by marked apathy and indifference. The individual may have no interest in his or her usual hobbies and appear unconcerned with events occurring in the immediate environment. (Both of these patterns may be associated with damage to the frontal lobes, and for this reason are sometimes referred to as "frontal lobe syndromes.") Another recognized pattern, seen in some individuals with temporal lobe epilepsy, is a marked tendency to humorless verbosity in both writing and speech, religiosity, and, occasionally, exaggerated aggressiveness.

The major personality change may be the development of suspiciousness or paranoid ideation.

Associated features. Mild cognitive impairment and irritability may be present.

Course. The course depends on the etiology. It may be transient, if it is the result of a medication or other substance use, or persistent, if it is secondary to

structural damage to the brain. Occasionally an Organic Personality Syndrome is the first manifestation of a disease process that eventually causes a Dementia. For example, in multiple sclerosis an Organic Personality Syndrome may precede the eventual development of a Dementia.

Impairment. The degree of impairment is variable. Despite relatively intact cognitive function, poor judgment may lead to such difficulties that the person may require constant supervision or even custodial care.

Complications. Socially unacceptable behavior may lead to social ostracism or legal difficulties. Impulsive or explosive behavior may be dangerous to the individual and to others.

Etiological factors. Organic Personality Syndrome is usually due to structural damage to the brain. The most common causes are neoplasms (for example, meningiomas pressing on the frontal lobes), head trauma (including the post-concussion syndrome), and vascular disease. A characteristic Organic Personality Syndrome has been described as an interictal phenomenon in some individuals with temporal lobe epilepsy. Multiple sclerosis and Huntington's chorea are sometimes associated with this syndrome. Rather uncommon causes are endocrine disorders (thyroid and adrenocortical disease) and the ingestion of certain substances.

Differential diagnosis. In **Dementia**, personality change is but one facet of an overall syndrome that also includes significant loss of intellectual abilities. Occasionally, personality change may be the first sign of an organic brain syndrome that will eventually evolve into Dementia. In these instances the initial diagnosis of Organic Personality Syndrome will have to be changed to Dementia as intellectual deficits increase and become the predominant feature. In **Organic Affective Syndrome** there may be a personality change, but a mood disturbance is the predominant clinical feature.

When **Attention Deficit Disorder** develops in a child or adolescent and is due to a specific organic factor, such as a known neurological disease, the additional diagnosis of Organic Personality Syndrome should not be made if the disturbance is limited to an impairment of impulse control and attention.

In **Schizophrenia**, **Paranoid Disorders**, **Affective Disorders**, and **Disorders of Impulse Control Not Elsewhere Classified**, marked personality changes may occur. In these disorders, however, no specific organic factor is judged etiologically related to the personality change.

Diagnostic criteria for Organic Personality Syndrome

A. A marked change in behavior or personality involving at least one of the following:

- (1) emotional lability, e.g., explosive temper outbursts, sudden crying

- (2) impairment in impulse control, e.g., poor social judgment, sexual indiscretions, shoplifting
- (3) marked apathy and indifference, e.g., no interest in usual hobbies
- (4) suspiciousness or paranoid ideation

B. No clouding of consciousness, as in Delirium; no significant loss of intellectual abilities, as in Dementia; no predominant disturbance of mood, as in Organic Affective Syndrome; no predominant delusions or hallucinations, as in Organic Delusional Syndrome or Organic Hallucinosi.

C. Evidence, from the history, physical examination, or laboratory tests, of a specific organic factor that is judged to be etiologically related to the disturbance.

D. This diagnosis is not given to a child or adolescent if the clinical picture is limited to the features that characterize Attention Deficit Disorder (see p. 41).

Intoxication

The essential features are maladaptive behavior and a substance-specific syndrome that are due to the recent use and presence in the body of a substance. Evidence for the recent use and presence in the body of the substance can be obtained by history, physical examination (e.g., smell of alcohol), or laboratory tests of urine or blood. The clinical picture does not correspond to any of the specific Organic Brain Syndromes, such as Delirium, Organic Delusional Syndrome, Organic Hallucinosi, or Organic Affective Syndrome. (Intoxication may, however, be superimposed on any Organic Brain Syndrome, with the exception of Delirium.)

As used here, the concept of Intoxication is a residual category for a clinical picture, caused by an exogenous substance, that does not correspond to any of the specific Organic Brain Syndromes. Thus, use of amphetamine that caused clouding of consciousness and tactile hallucinations would be diagnosed as Amphetamine Delirium, and only the milder clinical picture, without the full symptomatic picture of Delirium, would be diagnosed as Amphetamine Intoxication. Similar states caused by endogenous substances (such as ketone bodies in diabetic acidosis) should be coded in the category Atypical or Mixed Organic Brain Syndrome (see p. 162).

Although the specific clinical picture is determined by the nature of the substance used, the most common changes involve disturbances of perception, wakefulness, attention, thinking, judgment, emotional control, and psychomotor behavior.

As used here, Intoxication refers to a mental disorder and requires the presence of maladaptive behavior. It should be noted that when used in the physiological sense, the term "intoxication" is broader in scope than this definition

implies. Therefore, recreational use of substances that causes physiological and psychological changes but that does not result in maladaptive behavior is not considered Intoxication. For example, social drinking frequently causes loquacity, euphoria, and slurred speech; but this should not be considered Intoxication unless maladaptive behavior, such as fighting, impaired judgment, interference with social functioning, poor job performance, or failure to meet responsibilities, results.

Associated features. The particular substance used, the individual's expectations, pre-intoxication personality and biological state, and the environmental circumstances of the substance use determine the associated features. For example, Cannabis Intoxication may be intensified in an individual expecting religious revelations from the substance. Intoxication in public places may result in belligerent behavior or accidents. Withdrawn individuals may use intoxication as a method of achieving further social withdrawal (as in the "solitary drinker"). Individuals with depression may become suicidal or experience increased dysphoria when intoxicated.

Course. The rapidity of onset and the duration of an Intoxication depend on: the amount of the substance consumed, how rapidly it was consumed, the individual's tolerance (innate or acquired), body size (in general, the larger the person, the larger the quantity required to induce intoxication), and the half-life of the particular substance. In the case of alcohol, other considerations are the specific beverage consumed (distilled spirits produce higher blood levels than the same amount of alcohol consumed in the same period in the form of beer or wine) and whether the alcohol is combined with food.

Intoxication usually lasts for a few hours, but may last several days. In rare instances an individual may continue taking the substance so that he or she remains intoxicated for even longer periods.

Impairment. During the intoxicated state the individual has, by definition, some impairment in social or occupational functioning. This impairment in ability to function may be only minimal—for example, in Cannabis Intoxication at a time when the individual knows that he or she has no social or occupational responsibilities. On the other hand, impairment may be marked, for example, in Alcohol Intoxication, if the intoxication occurs when the individual has major social or occupational responsibilities.

Complications. Excessive ingestion of substances that have a depressant effect on the nervous system may result in coma or death. Substances that act as stimulants may cause seizures. During the intoxicated state the individual may fall or become involved in an automobile accident.

An individual who has repeated instances of Intoxication over a period of one month also has, by definition, Substance Abuse or Dependence. For some of the substances, an initial Intoxication may develop into a specific Organic Brain Syndrome. For example, Amphetamine Intoxication may develop into Amphetamine Delirium; or methyl alcohol Intoxication may lead to a Dementia.

Differential diagnosis. When the criteria for **Delirium**, **Organic Hallucinosiis**, **Organic Delusional Syndrome** or **Organic Affective Syndrome** are met following the ingestion of a substance, these conditions are diagnosed, because Intoxication is a residual category.

Many **neurological diseases** can produce symptoms, such as slurred speech and incoordination, that resemble an Intoxication.

Diagnostic criteria for Intoxication

A. Development of a substance-specific syndrome that follows the recent ingestion and presence in the body of a substance.

B. Maladaptive behavior during the waking state due to the effect of the substance on the central nervous system, e.g., impaired judgment, belligerence.

C. The clinical picture does not correspond to any of the specific Organic Brain Syndromes, such as Delirium, Organic Delusional Syndrome, Organic Hallucinosiis, or Organic Affective Syndrome.

Withdrawal

The essential feature is the development of a substance-specific syndrome that follows the cessation of or reduction in intake of a substance that was previously regularly used by the individual to induce a physiological state of intoxication.* Evidence for the cessation or reduction of regular use of a substance may be obtained by history or by laboratory tests of urine or blood. The clinical picture does not correspond to any of the specific Organic Brain Syndromes, such as Delirium, Organic Hallucinosiis, Organic Delusional Syndrome, and Organic Affective Syndrome. (Withdrawal may, however, be superimposed on any Organic Brain Syndrome with the exception of Delirium.)

As used here, the concept of Withdrawal is a residual category for a clinical picture caused by cessation of or reduction in intake of a substance that does not correspond to any of the specific Organic Brain Syndromes. Thus, a clinical syndrome of clouding of consciousness and tactile hallucinations following cessation of alcohol use is diagnosed as Alcohol Withdrawal Delirium, and only the milder clinical picture associated with cessation of alcohol use is termed Alcohol Withdrawal.

The syndrome that develops varies according to the substance the individual was using. Common symptoms include anxiety, restlessness, irritability, insomnia, and impaired attention.

Associated features. The nature of the substance determines the associated features, which may range from disturbing physiological symptoms, such as nausea and vomiting following cessation of heavy alcohol intake, to diffuse ma-

* Strictly speaking, in the case of the fetal alcohol syndrome, it is the mother, and not the infant, who has ingested the substance for the purpose of inducing intoxication.

laise, such as that following chronic use of certain sedatives, or compelling desire to resume taking the substance. There may be changes in sleep patterns and mood, as can be seen after withdrawal from amphetamines or corticosteroids, or convulsions after Barbiturate Withdrawal.

Course. Withdrawal is usually self-limited to no more than a few days or, at most, several weeks, except when complicated by the development of a specific Organic Brain Syndrome, such as a Delirium.

Impairment. Impairment varies from mild, as in Tobacco Withdrawal, to severe, as may be the case in Alcohol or Opioid Withdrawal.

Complications. In order to avoid painful withdrawal symptoms, the individual may continue to use the substance. Illegal behavior, such as stealing to pay for heroin, may result from efforts to obtain the substance.

Differential diagnosis. Many physical disorders have symptoms that are similar to the symptoms of Withdrawal. For example, the symptoms of influenza are very similar to the symptoms of Opioid Withdrawal.

Diagnostic criteria for Withdrawal

- A. Development of a substance-specific syndrome that follows the cessation of or reduction in intake of a substance that was previously regularly used by the individual to induce a state of intoxication.
- B. The clinical picture does not correspond to any of the specific Organic Brain Syndromes, such as Delirium, Organic Delusional Syndrome, Organic Hallucinoses, or Organic Affective Syndrome.

Atypical or mixed organic brain syndrome

This is a residual category reserved for syndromes that do not meet the criteria for any of the other Organic Brain Syndromes and in which there are maladaptive changes during the waking state with evidence, from either physical examination, laboratory tests, or history, of a specific organic factor that is judged to be etiologically related to the disturbance. An example would be the "neurasthenic" picture associated with early Addison's Disease.

This category should also be used when the individual has an Organic Brain Syndrome with features of more than one syndrome but not enough of any one to meet the criteria for a single syndrome.

Diagnostic criteria for Atypical or Mixed Organic Brain Syndrome

- A. The disturbance occurs during the waking state and does not fulfill the criteria for any of the previously described Organic Brain Syndromes.

B. Evidence, from the history, physical examination, or laboratory tests, of a specific organic factor that is judged to be etiologically related to the disturbance.

ORGANIC MENTAL DISORDERS—SECTION 1

Organic Mental Disorders in which the etiology or pathophysiological process is listed below (taken from the mental disorders section of ICD-9-CM).

This section consists of two groups of disorders of organic etiology that are traditionally classified as mental disorders: the first category includes Dementias due to certain neurological diseases characteristically appearing in the senium and presenium; the second is composed of Substance-induced Organic Mental Disorders.

DEMENTIAS ARISING IN THE SENIUM AND PRESENIUM

The Dementias associated with Alzheimer's and Pick's diseases have been referred to as Senile and Presenile Dementias, the former arbitrarily signifying an age at onset over 65. Since nearly all cases of these Dementias are associated with Alzheimer's disease and the identification of Alzheimer's and Pick's diseases is largely or entirely dependent on histopathological data, it seems more useful to have in a clinical classification of mental disorders a single category that encompasses the syndrome of Primary Degenerative Dementia. This category is subtyped according to the age at onset, for the purpose of historical continuity and to maintain comparability with ICD-9-CM. The clinician will rarely be in a position to identify the specific associated neurological disorder. When such information is available, it should be noted on Axis III.

In DSM-II, the Dementia associated with vascular disease was called Psychosis with Cerebral Arteriosclerosis. However, the severity of the disorder appears to be related to repeated infarcts of the brain rather than to the extent of cerebral arteriosclerosis. At autopsy the brain shows multiple infarcts of various ages. For this reason, this category is here termed Multi-infarct Dementia.

When a Dementia is due to some other known disease, such as a brain tumor, Huntington's chorea, or vitamin B-12 deficiency, the specific disease should be noted on Axis III, and the presence of a Dementia, on Axis I (294.10 Dementia, from Section 2 of this chapter, p. 162).

290.xx Primary Degenerative Dementia

The essential feature is the presence of Dementia of insidious onset and gradually progressive course for which all other specific causes have been excluded by the history, physical examination, and laboratory tests. The Dementia involves a multifaceted loss of intellectual abilities, such as memory, judgment, abstract thought, and other higher cortical functions, and changes in personality and behavior. (See p. 107 for a description of the essential and associated features of Dementia.)

Phenomenological subtypes. Frequently the clinical picture may be compli-

cated by the presence of significant depressive features or of delusions, usually persecutory. More rarely there may be a superimposed Delirium. These additional features, when present, should be noted with the appropriate codes (p. 126).

Age at onset. Senile onset (after age 65) is much more common than pre-senile onset. Few cases develop before the age of 49.

Course. The onset is insidious, and the course is one of uniform, gradual progression. In the early stages memory impairment may be the only apparent cognitive deficit. There may also be subtle personality changes, such as the development of apathy, lack of spontaneity, and a quiet withdrawal from social interactions. Individuals usually remain neat and well-groomed and, aside from an occasional irritable outburst, are cooperative and behave in a socially appropriate way. With progression to the middle stage of the disease, various cognitive disturbances become quite apparent, and behavior and personality are more obviously affected. By the late stage, the individual may be completely mute and inattentive. At this point he or she is totally incapable of caring for himself or herself. This stage leads inevitably to death. With senile onset, the average duration of symptoms, from onset to death, is about five years.

Impairment and complications. See Dementia, p. 110.

Pathology. In the majority of cases, the brain is atrophied, with widened cortical sulci and enlarged cerebral ventricles. This may be demonstrated in life by computer-assisted tomography or pneumoencephalography. Microscopic examination usually reveals three histopathological changes: senile plaques, neurofibrillary tangles, and granulovacuolar degeneration of neurons. These are the changes classically described in Alzheimer's disease. Rare cases have the histological features of Pick's disease, mixed vascular and degenerative disease, or nonspecific pathological changes.

Predisposing factors. Down's syndrome predisposes to Alzheimer's disease.

Prevalence. Between 2% and 4% of the population over the age of 65 is estimated to have Primary Degenerative Dementia. The prevalence increases with increasing age, particularly after 75.

Sex ratio. The disorder is more common in women than in men.

Familial pattern. First-degree relatives are four times more likely to develop the disease than members of the general population. In rare cases, Primary Degenerative Dementia of the Alzheimer's type is inherited as a dominant trait.

Differential diagnosis. The normal process of aging has been associated in a number of studies with certain decrements in intellectual functioning. The nature and significance of these changes are controversial. The diagnosis of Pri-

mary Degenerative Dementia should be limited to cases in which there is clear evidence of progressive and significant deterioration of intellectual and social or occupational functioning. (See Dementia, differential diagnosis, p. 110, for discussion.)

Subdural hematoma, normal-pressure hydrocephalus, cerebral neoplasm, Parkinson's disease, vitamin B₁₂ deficiency, hypothyroidism, substance intoxication, and other specific and possibly treatable physical disorders that may cause Dementia need to be ruled out by the history, physical examination, and appropriate laboratory tests.

In **Multi-infarct Dementia** the clinical course is more variable and typically progresses in stepwise fashion with focal neurological signs and systemic evidence of vascular disease. In occasional cases, the two disorders may coexist, and both diagnoses should be recorded.

Elderly individuals with a major depressive episode may have features strongly suggesting Dementia. For a discussion of this important problem in differential diagnosis, see **major depressive episode** (p. 212) and **Dementia** (p. 111).

Subtypes

Primary Degenerative Dementia, Senile Onset (after age 65)

290.30 with delirium

290.20 with delusions

290.21 with depression

290.00 uncomplicated

Primary Degenerative Dementia, Presenile Onset (age 65 and below)

290.11 with delirium

290.12 with delusions

290.13 with depression

290.10 uncomplicated

Diagnostic criteria for Primary Degenerative Dementia

A. Dementia (see p. 111).

B. Insidious onset with uniformly progressive deteriorating course.

C. Exclusion of all other specific causes of Dementia by the history, physical examination, and laboratory tests.

290.4x Multi-infarct Dementia

The essential feature of this Dementia is a stepwise deterioration in intellectual functioning that early in the course leaves some intellectual functions relatively intact ("patchy" deterioration).

Focal neurological signs and symptoms are also present, and there is evidence of significant cerebrovascular disease that is judged to be etiologically related to the disturbance.

The onset is typically abrupt; and the course is stepwise and fluctuating, with rapid changes, rather than uniformly progressive. The pattern of deficits is "patchy," depending upon which regions of the brain have been destroyed. Certain cognitive functions may be affected early while others remain relatively unimpaired. The Dementia typically involves disturbances in memory, abstract thinking, judgment, impulse control, and personality. See Dementia, p. 107.

The focal neurological signs commonly seen include weaknesses in the limbs, reflex asymmetries, extensor plantar responses, dysarthria, and small stepped gait.

Vascular disease is always presumed to be present and responsible for both the Dementia and the focal neurological signs. Evidence of cerebral and systemic vascular disease may be apparent on physical examination. Frequently there will be hypertension, carotid bruit, funduscopic abnormalities, or an enlarged heart. These conditions should be noted on Axis III.

Associated features. Pseudobulbar palsy, with fleeting episodes of laughing and crying (sham emotion), dysarthria, and dysphagia is very common. There may be periods of increased confusion, possibly related to further vascular insults.

Age at onset. The onset of Multi-infarct Dementia is apparently earlier than that of Primary Degenerative Dementia.

Course. As noted above, the course is erratic. Early treatment of hypertension and vascular disease may prevent further progression.

Pathology. The brain shows multiple and extensive localized areas of softening. There may also be a variety of pathological changes in the cerebral vessels.

Predisposing factors. The most significant factor is arterial hypertension. Extracranial vascular disease of the great vessels in the neck and valvular disease of the heart may be sources of cerebral emboli.

Prevalence. Multi-infarct Dementia is apparently much less common than Primary Degenerative Dementia.

Sex ratio. The disorder is apparently more common in men.

Familial pattern. No information.

Differential diagnosis. A single stroke may cause a relatively circumscribed change in the mental state, such as an aphasia following damage to the left hemisphere, or an Amnesic Syndrome from infarction in the region of the posterior cerebral arteries. As a general rule, a single stroke does not cause Dementia. Multi-infarct Dementia results from the occurrence of multiple strokes, at different times.

In **Primary Degenerative Dementia** the course is uniformly progressive rather than stepwise as in Multi-infarct Dementia, and there is usually no evidence of cerebrovascular disease. In some instances both Multi-infarct Dementia and Primary Degenerative Dementia may coexist, with clinical features of both entities. In such cases both diagnoses should be recorded.

Subtypes

Multi-infarct Dementia

290.41 with delirium

290.42 with delusions

290.43 with depression

290.40 uncomplicated

Diagnostic criteria for Multi-infarct Dementia

A. Dementia (see p. 111).

B. Stepwise deteriorating course (i.e., not uniformly progressive) with "patchy" distribution of deficits (i.e., affecting some functions, but not others) early in the course.

C. Focal neurological signs and symptoms (e.g., exaggeration of deep tendon reflexes, extensor plantar response, pseudobulbar palsy, gait abnormalities, weakness of an extremity, etc.).

D. Evidence, from the history, physical examination, or laboratory tests, of significant cerebrovascular disease that is judged to be etiologically related to the disturbance.

SUBSTANCE-INDUCED ORGANIC MENTAL DISORDERS*

This section of the classification deals with the various Organic Brain Syndromes caused by the direct effects on the nervous system of various substances. They are distinguished from the Substance Use Disorders that refer to the behavior associated with taking substances that affect the central nervous system.

* In order for the DSM-III classification to be compatible with ICD-9-CM, some diagnoses in this subclass share the same code number. Non-ICD-9-CM codes are provided in parentheses in the classification (Chapter 1) for use when greater specificity is necessary.

In most cases the diagnosis of these Organic Mental Disorders will be made in individuals who also have a Substance Use Disorder.

This section includes those Substance-induced Organic Mental Disorders caused by the ten classes of substances that most commonly are taken nonmedically to alter mood or behavior: alcohol, barbiturates or similarly acting sedatives or hypnotics, opioids, cocaine, amphetamines or similarly acting sympathomimetics, phencyclidine (PCP) or similarly acting arylcyclohexylamines, hallucinogens, cannabis, tobacco, and caffeine. Although some of these substances also have a legitimate medicinal purpose, they may under unsupervised circumstances cause Organic Mental Disorders. In addition, there is a residual class for Organic Mental Disorders caused by other or unknown substances.

For each class of substances, the specific disorders described represent the types of Organic Brain Syndrome known to be caused by that class of substance. For example, whereas alcohol causes many different syndromes, the current evidence suggests that cocaine causes only one.

The descriptions of these disorders often do not include many categories of information, such as age at onset, predisposing factors, prevalence, and sex ratio. This information is frequently available in the corresponding portion of the Substance Use Disorder section. The description of Impairment for each of the individual Organic Brain Syndromes can be found in the preceding section. For many of the Substance-induced Organic Mental Disorders, a predisposing factor is prolonged heavy use with the development of dependence. Since this is obvious, only other predisposing factors are noted, when known.

Recording specific diagnoses. Whenever possible the clinician should record the name of the specific substance rather than the name of the entire class of substances, using the code number for the appropriate class. Examples: the clinician should write 305.70 Amphetamine Intoxication (rather than Amphetamine or Similarly Acting Sympathomimetic Intoxication); 292.00 Valium Withdrawal (rather than Barbiturate or Similarly Acting Sedative or Hypnotic Withdrawal); 292.81 Atropine Delirium (rather than Other or Unspecified Substance Delirium).

ALCOHOL ORGANIC MENTAL DISORDERS

This section includes the following Organic Mental Disorders attributed to the ingestion of alcohol: Alcohol Intoxication, Alcohol Idiosyncratic Intoxication, Alcohol Withdrawal, Alcohol Withdrawal Delirium, Alcohol Hallucinoses, Alcohol Amnestic Disorder and Dementia Associated with Alcoholism.

Although ICD-9-CM has a category for Alcoholic Jealousy, the literature does not provide sufficient evidence to support the existence of this syndrome as an independent entity. The concept "alcoholic jealousy" can be expressed in DSM-III terms by a diagnosis of Alcohol Dependence and an additional diagnosis of a Paranoid Disorder.

303.00 Alcohol Intoxication

The essential feature is maladaptive behavior due to the recent ingestion of

alcohol. This may include aggressiveness, impaired judgment, and other manifestations of impaired social or occupational functioning. Characteristic physiological signs include flushed face, slurred speech, unsteady gait, nystagmus, and incoordination. Characteristic psychological signs include loquacity, impaired attention, irritability, euphoria, depression, and emotional lability.

The diagnosis is not made when there is evidence that the amount of alcohol ingested was insufficient to cause intoxication in most people, as in Alcohol Idiosyncratic Intoxication (p. 132).

Associated features. The individual's usual behavior may be accentuated or altered. For example, an individual who tends to be somewhat suspicious may, under the influence of alcohol, become markedly paranoid. On the other hand, individuals who are ordinarily withdrawn and uncomfortable in social situations may become exceptionally convivial.

Alcohol Intoxication is sometimes associated with an amnesia for the events that occurred during the course of the intoxication ("blackouts").

Course. Although alcohol is basically a central nervous system depressant, its initial behavioral effects are often viewed as "disinhibitory" phenomena. Thus, early in the course of Alcohol Intoxication, an individual may appear exceptionally bright, expansive, and hyperactive, with a subjective sensation of well-being and increased mental sharpness. With further intoxication, however, the individual may slow down and become depressed, withdrawn, and dull, and even lose consciousness.

The duration of an episode of Alcohol Intoxication depends on a variety of factors, including the amount and type of alcoholic beverage consumed, how rapidly it was ingested, and whether or not it was taken with food. Since only a small percentage of alcohol is excreted, the rate at which alcohol is metabolized—approximately 5-10 ml per hour—plays a very important role in determining the length of a period of intoxication. The development of short-term tolerance may also influence the course of Alcohol Intoxication so that a person may appear less intoxicated after many hours of drinking than after a few hours. There is considerable individual variation in susceptibility to intoxication with alcohol. Some individuals show intoxication with blood alcohol levels as low as 30 mg % whereas others appear unintoxicated with levels as high as 150 mg %.

The signs of intoxication are more marked when the blood alcohol level is rising than when it is falling. Most individuals become intoxicated at blood alcohol levels between 100 and 200 mg %. Death has been reported with levels ranging from 400 to 700 mg %. Alcohol exerts its fatal effect either by a direct depression of respiration or by aspiration of vomitus.

Complications. Highway accidents are a major complication of Alcohol Intoxication. At least half of all highway fatalities involve either a driver or a pedestrian who has been drinking. Intoxication also results in falls and numerous household and industrial accidents. Moreover, it is frequently associated with the commission of criminal acts. More than one-half of all murderers and their victims are believed to be intoxicated at the time of the act. One study in-

indicates that about one-fourth of all suicides occur while the individual is drinking alcohol.

Alcohol Intoxication frequently results in physical disorders. Falls and accidents result in fractures, subdural hematomas, and other forms of brain trauma. Exposure to extreme weather leads to frostbite and sunburn. Alcohol Intoxication may also possibly suppress immune mechanisms and thus predispose to infections.

Differential diagnosis. Social drinking is associated with physiological intoxication. Maladaptive behavior is required for the mental disorder Alcohol Intoxication to be diagnosed.

Intoxication due to barbiturates and similarly acting sedatives and hypnotics has the same clinical picture as Alcohol Intoxication. Since an individual may be taking both alcohol and other substances, the presence of alcohol on the breath does not exclude the possibility that another substance is responsible for the intoxication.

Certain neurological diseases, such as **cerebellar ataxias** or **multiple sclerosis**, may have some of the physiological signs and symptoms of Alcohol Intoxication.

In **Alcohol Idiosyncratic Intoxication**, a marked change in behavior follows ingestion of an amount of alcohol that is insufficient to cause Alcohol Intoxication in most individuals.

Diagnostic criteria for Alcohol Intoxication

A. Recent ingestion of alcohol (with no evidence suggesting that the amount was insufficient to cause intoxication in most people).

B. Maladaptive behavioral effects, e.g., fighting, impaired judgment, interference with social or occupational functioning.

C. At least one of the following physiological signs:

- (1) slurred speech
- (2) incoordination
- (3) unsteady gait
- (4) nystagmus
- (5) flushed face

D. At least one of the following psychological signs:

- (1) mood change
- (2) irritability
- (3) loquacity
- (4) impaired attention

E. Not due to any other physical or mental disorder.

291.40 Alcohol Idiosyncratic Intoxication

The essential feature is a marked behavioral change—usually to aggressiveness—that is due to the recent ingestion of an amount of alcohol insufficient to induce intoxication in most people. There is usually subsequent amnesia for the period of intoxication. The behavior is atypical of the person when not drinking—for example, a shy, retiring, mild-mannered person may, after one weak drink, become belligerent and assaultive. During the episode the individual seems out of contact with others.

This disorder has also been called “Pathological Intoxication.”

Age at onset. No information.

Course. The change in behavior begins either while the individual is drinking or shortly thereafter. The duration is quite brief, and the condition ceases within a few hours. The individual returns to his or her normal state as the blood alcohol level falls.

Prevalence. Apparently extremely rare.

Complications. The individual may do harm to himself or herself or to others.

Predisposing factors. A small percentage of individuals with this disorder have been reported to have temporal lobe spikes on an electroencephalogram after receiving small amounts of alcohol. Although the reports are still anecdotal, it is thought that people with brain damage lose “tolerance” for alcohol and behave abnormally after drinking small amounts. The types of brain injury most often associated with this syndrome are from trauma and encephalitis. The loss of tolerance may be temporary or permanent. It is also reported that individuals who are unusually fatigued or have a debilitating physical illness may have a low tolerance for alcohol and respond inappropriately to small amounts.

Differential diagnosis. Other exogenous agents, especially barbiturates and similarly acting substances, may occasionally cause abrupt changes in behavior. Temporal lobe epilepsy, during the interictal period, may be associated with fits of destructive rage. In *Malingering*, the individual may wish to avoid responsibility for aggressive behavior, claiming that it occurred while he or she was intoxicated from a small amount of alcohol.

Diagnostic criteria for Alcohol Idiosyncratic Intoxication

A. Marked behavioral change, e.g., aggressive or assaultive behavior that is due to the recent ingestion of an amount of alcohol insufficient to induce intoxication in most people.

B. The behavior is atypical of the person when not drinking.

C. Not due to any other physical or mental disorder.

291.80 Alcohol Withdrawal

The essential features are certain characteristic symptoms such as a coarse tremor of the hands, tongue, and eyelids, nausea and vomiting, malaise or weakness, autonomic hyperactivity (such as tachycardia, sweating, and elevated blood pressure), anxiety, depressed mood or irritability, and orthostatic hypotension, that follow within several hours cessation of or reduction in alcohol ingestion by an individual who has been drinking alcohol for several days or longer. The diagnosis is not made if the disturbance is Alcohol Withdrawal Delirium.

Associated features. Headache and dry mouth, not necessarily due to dehydration, are frequent symptoms. The complexion is often puffy and blotchy, and there may be mild peripheral edema. When nausea and vomiting are present, there may also be gastritis.

Sleep is often fitful and disturbed by "bad dreams." These merge with a variety of misperceptions and illusions. Brief, poorly formed hallucinations, occurring in any modality of sensation, may be experienced.

Age at onset. Most individuals with Alcohol Dependence begin drinking early in life and develop their first withdrawal symptoms in their 30s or 40s.

Course. Withdrawal symptoms begin shortly after cessation of or reduction in drinking and almost always disappear within five to seven days, unless Alcohol Withdrawal Delirium develops.

Complications. Major motor seizures ("rum fits") may occur. Individuals with a preexisting history of epilepsy are more apt to develop withdrawal seizures.

Predisposing factors. Malnutrition, fatigue, depression, and concomitant physical illness may aggravate the syndrome.

Differential diagnosis. In Alcohol Withdrawal Delirium there are a clouded state of consciousness and other symptoms characteristic of Delirium.

In Alcohol Hallucinosis the hallucinations are prominent and persistent whereas if they occur in Alcohol Withdrawal, they are brief and poorly formed.

Barbiturate or Similarly Acting Sedative or Hypnotic Withdrawal produces a syndrome essentially identical with that of Alcohol Withdrawal. **Hypoglycemia**, whether or not related to alcohol, and **diabetic ketoacidosis** result in symptoms similar to those of Alcohol Withdrawal. **Essential tremor** may suggest the tremulousness of Alcohol Withdrawal.

Diagnostic criteria for Alcohol Withdrawal

A. Cessation of or reduction in heavy prolonged (several days or longer) ingestion of alcohol, followed within several hours by coarse tremor of hands, tongue, and eyelids and at least one of the following:

- (1) nausea and vomiting
- (2) malaise or weakness
- (3) autonomic hyperactivity, e.g., tachycardia, sweating, elevated blood pressure
- (4) anxiety
- (5) depressed mood or irritability
- (6) orthostatic hypotension

B. Not due to any other physical or mental disorder such as Alcohol Withdrawal Delirium.

291.00 Alcohol Withdrawal Delirium

The essential feature is a Delirium (p. 104) that is due to recent cessation of or reduction in alcohol consumption. Autonomic hyperactivity, such as tachycardia and sweating, and elevated blood pressure, is present. Delusions, vivid hallucinations, and agitated behavior usually occur. Hallucinations, when present, are usually visual, but may occur in other sensory modalities.

This disorder has been called "delirium tremens."

Associated features. A coarse, irregular tremor is almost always present. Fever may also occur.

Age at onset. The first episode of this disorder usually occurs after 5 to 15 years of heavy drinking, usually episodic. For this reason, the disorder usually first occurs in the person's 30s or 40s.

Course. The onset is usually on the second or third day after the cessation of or reduction in drinking. Occasionally it occurs earlier; it rarely appears more than a week after abstinence. Unless complicated by some other illness, the syndrome most often runs its course in two to three days. If seizures also occur as the result of Alcohol Withdrawal ("rum fits"), they always precede the development of Delirium.

Complications. See Delirium (p. 106).

Predisposing factors. The presence of a concomitant physical illness may predispose to this syndrome.

Prevalence. This syndrome is much less common than Alcohol Withdrawal. In one study, no more than 5% of patients admitted to a general hospital for Alcohol Dependence had Alcohol Withdrawal Delirium.

Differential diagnosis. See Delirium (p. 106).

Diagnostic criteria for Alcohol Withdrawal Delirium

- A. Delirium (p. 107) occurs within one week after cessation of or reduction in heavy alcohol ingestion.
- B. Autonomic hyperactivity, e.g., tachycardia, sweating, elevated blood pressure.
- C. Not due to any other physical or mental disorder.

291.30 Alcohol Hallucinosis

The essential feature is an Organic Hallucinosis (see p. 115) with vivid auditory hallucinations following cessation of or reduction in alcohol ingestion by an individual who apparently has Alcohol Dependence. The hallucinations are usually voices, and less commonly unformed sounds such as hissing or buzzing. Onset may accompany a gradual decrease in blood alcohol levels toward the end of an extended period of Intoxication; but it most often occurs soon after cessation of drinking, usually within the first 48 hours, although occasionally somewhat later.

In the majority of cases, the content of the hallucinations is unpleasant and disturbing. However, the hallucinatory content may be benign, leaving the individual undisturbed. The voices may address the individual directly, but more often they discuss him or her in the third person. When the voices are threatening, the individual may act to defend himself or herself by calling on the police for protection, or arming against invaders. The actions of the individual are practically never the result of the hallucinations' commanding the individual to act in a certain way, but rather are motivated by the desire to avoid disgrace, injury, or other consequences of what the voices threaten.

Associated features. Other signs of withdrawal following prolonged intoxication may occur, such as transient visual hallucinations, tremulousness, seizures, and, in some cases, Delirium.

Age at onset. Although first episodes have been reported in people in their early to mid-20s, the more typical onset is about age 40, and follows 10 years or more of heavy drinking.

Course. Most frequently the disorder lasts only a few hours or days, typically, less than a week. In about 10% of cases, however, it may last several weeks or months; in a few cases, a chronic form develops.

With the evolution of the chronic form, which may be recognized as early as a week after the onset of the illness, the individual becomes quiet and resigned despite the fact that the hallucinations remain threatening and derogatory. Ideas of reference and other poorly systematized persecutory delusions become prominent. At this stage the illness may be clinically indistinguishable from Schizophrenia, with vague and illogical thinking, tangential associations, and inappropriate affect. There is some evidence that the chronic form is more likely to develop from repeated episodes of the disorder.

Impairment. Impairment may be severe, as the individual responds to the hallucinations as though they were real.

Complications. In an effort to avoid the consequences of threatening voices, the individual may harm himself or herself or others.

Predisposing factors. This disorder occurs only following prolonged, heavy ingestion of alcohol in individuals who apparently have Alcohol Dependence. Contrary to previously held beliefs, there is no evidence that Schizophrenia predisposes to the development of this disorder.

Prevalence. Apparently rare.

Sex ratio. The disorder is apparently four times more common in males than in females.

Familial pattern. No information.

Differential diagnosis. In **Schizophrenia** there is no temporal relation of psychotic symptoms to the cessation of alcohol use and there is a chronic course. Moreover, the age at onset of Alcohol Hallucinosis is later than in Schizophrenia, and both the family background and pre-illness personalities are not typical of those for Schizophrenia.

Diagnostic criteria for Alcohol Hallucinosis

A. Organic Hallucinosis (see p. 116) with vivid auditory hallucinations developing shortly (usually within 48 hours) after cessation of or reduction in heavy ingestion of alcohol in an individual who apparently has Alcohol Dependence.

B. Response to the hallucinations appropriate to their content, e.g., anxiety in response to hallucinatory threats.

C. No clouding of consciousness, as in Delirium.

D. Not due to any other physical or mental disorder.

291.10 Alcohol Amnestic Disorder

The essential feature is an Amnestic Syndrome (see p. 112) due to the vitamin deficiency associated with prolonged, heavy use of alcohol. Alcohol Amnestic Disorder due to thiamine deficiency is also known as Korsakoff's Disease.

Associated features. Neurological disturbances such as peripheral neuropathy, cerebellar ataxia, or myopathy are among the associated features.

Course. Alcohol Amnestic Disorder often follows an acute episode of Wernicke's encephalopathy, a neurological disease manifested by confusion, ataxia, eye-movement abnormalities (gaze palsies, nystagmus), and other neurological signs. Gradually these manifestations subside, but a major impairment of memory, Alcohol Amnestic Disorder, remains. If Wernicke's disease is treated early with large doses of thiamine, Alcohol Amnestic Disorder may not develop.

Once Alcohol Amnestic Disorder becomes established, it usually persists indefinitely. A slight degree of improvement over a long period of time may occur.

Impairment. Impairment is usually quite severe, and life-long custodial care may be necessary.

Complications. See Amnestic Syndrome, p. 113.

Predisposing factors. Alcohol Dependence (by definition).

Prevalence. The disorder is apparently rare, perhaps because of the routine administration of thiamine during detoxification.

Differential diagnosis. In **Dementia Associated with Alcoholism** the disturbance is not limited to memory impairment.

See Amnestic Syndrome, p. 113.

Diagnostic criteria for Alcohol Amnestic Disorder

A. Amnestic Syndrome (see p. 113) following prolonged heavy ingestion of alcohol.

B. Not due to any other physical or mental disorder.

291.2x Dementia Associated with Alcoholism

The essential feature is a Dementia (see p. 107) associated with prolonged and heavy ingestion of alcohol for which all other causes of Dementia have been excluded. In order to exclude transient effects of Intoxication and Withdrawal, this diagnosis should not be made until at least three weeks have elapsed since the cessation of alcohol use.

The etiologic role of alcohol in the Dementia associated with prolonged heavy ingestion of alcohol is controversial.

Subtyping. The severity of the Dementia is noted in the fifth digit (see below). The severe form corresponds to what was called Alcoholic Deterioration in DSM-II. Both the severe and the mild forms of this entity have been described by various studies in the literature. What has been termed "moderate" represents a theoretical intermediate stage. It is not established whether these forms repre-

sent different points along a continuum, or whether the mild and the severe forms have different pathophysiologies.

Associated features. See Alcohol Dependence (p. 169) and Dementia (p. 109).

Age at onset. This syndrome rarely begins before the age of 35 since many years of heavy drinking are apparently necessary to produce Dementia.

Course. No information.

Impairment. Mild cognitive deficits demonstrable only by psychological testing have been reported in some individuals with Alcoholism. As this diagnosis involves a Dementia, by definition there is always some impairment in social or occupational functioning. When impairment is severe, the individual becomes totally oblivious of his or her surroundings and requires constant care.

Complications. See Dementia, p. 110.

Predisposing factors. Alcohol Dependence (by definition).

Prevalence. Mild, sometimes reversible, intellectual impairment is commonly found when a specific search is made in individuals with chronic Alcohol Dependence. The severe form of this disorder is apparently rare.

Differential diagnosis. Dementia Associated with Alcoholism is distinguished from **Alcohol Amnestic Disorder** by the presence of cognitive deficits other than in the sphere of memory alone. It is distinguished from **other causes of Dementia** by the failure to demonstrate a specific etiology other than alcohol abuse. For the differential diagnosis of Dementia, see p. 110.

Diagnostic criteria for Dementia Associated with Alcoholism

A. Dementia (see p. 111) following prolonged, heavy ingestion of alcohol.

B. Dementia persisting at least three weeks after cessation of alcohol ingestion.

C. Exclusion of all other causes of Dementia, other than prolonged, heavy use of alcohol, by the history, physical examination, and laboratory tests.

Severity criteria for the Dementia

291.21 Mild

No more than mild impairment in social and occupational functioning.

291.22 Moderate

Moderate social impairment with inability to function occupationally.

291.23 Severe

Severe impairment of functioning with marked deterioration of personality (irritability, social inappropriateness) and inability to function independently.

291.20 Unspecified

BARBITURATE OR SIMILARLY ACTING SEDATIVE OR HYPNOTIC ORGANIC MENTAL DISORDERS

Included in this classification are disorders induced by barbiturates and similarly acting sedatives and hypnotics.

The common minor tranquilizers are the benzodiazepines, such as chlordiazepoxide, diazepam, and oxazepam. Common hypnotics include ethchlorvynol, flurazepam, glutethimide, methyprylon, chloral hydrate, paraldehyde, and methaqualone.

Although these substances differ widely in their rates of absorption, metabolism, distribution in the body, and likelihood of producing intoxication and withdrawal, at some dose and at some duration of use they are all capable of producing signs and symptoms of intoxication and withdrawal that are essentially the same as those produced by the barbiturates. For this reason the barbiturates are the prototype for this group of substances.

Substances in this class are usually taken orally in the form of pills or capsules.

Because of the great pharmacological differences in the substances covered in this section and of the wide individual variation in responses to them, specific doses for specific withdrawal syndromes are not listed.

305.40 Barbiturate or Similarly Acting Sedative or Hypnotic Intoxication

The essential and associated features are virtually identical with those of Alcohol Intoxication (see p. 129). The only exception is that there is no syndrome of Idiosyncratic Intoxication. Differences that may occur between Alcohol Intoxication and Intoxication by this class of substances are most likely due to differences in the personalities of the individuals who become intoxicated and in the settings in which the intoxications occur. For example, the settings in which Alcohol Intoxication is apt to occur probably account for the greater likelihood that it will be accompanied by displays of aggression or violence, compared with intoxication by this class of substances.

Course. As with Alcohol Intoxication, the initial behavioral effects are usually disinhibitory. If the individual continues to ingest the substance, inhibitory effects will supervene.

The factors involved in rapidity of onset and duration of the intoxication are discussed in the text on Intoxication as an Organic Brain Syndrome (see p. 121).

Complications. All of the complications noted for Alcohol Intoxication may occur with this class of substances, although they are less commonly seen. From initiation of regular barbiturate use to the development of physical dependence, the margin between the intoxicating dose and the lethal dose progressively narrows as the dose necessary for intoxication increases. The upper limit for the lethal dose usually does not exceed 3.5 to 4.0 grams. Therefore, a fatal overdose can be ingested accidentally. Unlike with Alcohol Intoxication, death is a more frequent complication because the individual is able to take enough of the substance over a short period of time to induce marked depression of central nervous system function.

Differential diagnosis. Other substances causing intoxication must be considered. Although breath odor may be an important differentiating clue with alcohol and many inhalants, it should not be absolutely relied upon.

Parenteral administration of naloxone or other opioid antagonists will alleviate the symptoms of **Opioid Intoxication**, but will have no effect on intoxications due to other substances.

Reliable qualitative and quantitative tests for the presence in the blood and urine of barbiturates and some of the other substances in this class may be useful.

Diagnostic criteria for Barbiturate or Similarly Acting Sedative or Hypnotic Intoxication

A. Recent use of a barbiturate or similarly acting sedative or hypnotic.

B. At least one of the following psychological signs:

- (1) mood lability
- (2) disinhibition of sexual and aggressive impulses
- (3) irritability
- (4) loquacity

C. At least one of the following neurological signs:

- (1) slurred speech
- (2) incoordination
- (3) unsteady gait
- (4) impairment in attention or memory

D. Maladaptive behavioral effects, e.g., impaired judgment, interference with social or occupational functioning, failure to meet responsibilities.

E. Not due to any other physical or mental disorder.

292.00 Barbiturate or Similarly Acting Sedative or Hypnotic Withdrawal

The essential features are virtually identical with those of Alcohol Withdrawal (see p. 133). The only exception is that a coarse tremor is not invariably present.

Diagnostic criteria for Barbiturate or Similarly Acting Sedative or Hypnotic Withdrawal

A. Prolonged, heavy use of barbiturate or similarly acting sedative or hypnotic, or more prolonged use of smaller doses of a benzodiazepine.

B. At least three of the following due to recent cessation of or reduction in substance use:

- (1) nausea and vomiting
- (2) malaise or weakness
- (3) autonomic hyperactivity, e.g., tachycardia, sweating, elevated blood pressure
- (4) anxiety
- (5) depressed mood or irritability
- (6) orthostatic hypotension
- (7) coarse tremor of hands, tongue, and eyelids

C. Not due to any other physical or mental disorder, such as Barbiturate or Similarly Acting Sedative or Hypnotic Withdrawal Delirium.

292.00 Barbiturate or Similarly Acting Sedative or Hypnotic Withdrawal Delirium

The essential feature is a Delirium (see p. 104) occurring within one week after cessation of or reduction in heavy use of a barbiturate or similarly acting sedative or hypnotic. All of the features are virtually identical with those of Alcohol Withdrawal Delirium (see p. 134), but the disorder is apparently not as common.

Diagnostic criteria for Barbiturate or Similarly Acting Sedative or Hypnotic Withdrawal Delirium

A. Delirium (see p. 107) within one week after cessation of or reduction in heavy use of a barbiturate or similarly acting sedative or hypnotic.

B. Autonomic hyperactivity, e.g., tachycardia, sweating, elevated blood pressure.

C. Not due to any other physical or mental disorder.

292.83 Barbiturate or Similarly Acting Sedative or Hypnotic Amnestic Disorder

The essential feature is an Amnestic Syndrome (see p. 112) due to prolonged heavy use of a barbiturate or similarly acting sedative or hypnotic.

Age at onset. There is some evidence that the age at onset is in the 20s. The earlier onset compared with that of Alcohol Amnestic Disorder may be due to the more common earlier age at onset of heavy use of this class of substances compared with alcohol.

Course. The course is variable. Unlike Alcohol Amnestic Disorder, full recovery may occur.

Impairment, complications, and differential diagnosis. See Amnestic Syndrome, p. 113.

Diagnostic criteria for Barbiturate or Similarly Acting Sedative or Hypnotic Amnestic Disorder

A. Prolonged, heavy use of a barbiturate or similarly acting sedative or hypnotic.

B. Amnestic Syndrome (see p. 113).

C. Not due to any other physical or mental disorder.

OPIOID ORGANIC MENTAL DISORDERS

This group includes natural opioids, such as heroin and morphine, and synthetics with morphine-like action, such as meperidine and methadone. These substances are taken either orally, intravenously, intranasally, or subcutaneously ("skin popping").

Although methadone is included in this class, individuals properly supervised in a methadone maintenance program should not develop any of the Opioid Organic Mental Disorders. When they do meet the criteria for this diagnosis, this indicates that there has been nonmedical use of methadone, in which case the appropriate diagnosis should be made.

305.50 Opioid Intoxication

The essential features are specific neurological and psychological signs and maladaptive behavioral effects due to the recent use of an opioid.

Psychological signs commonly present include euphoria or dysphoria, apathy, and psychomotor retardation.

Pupillary constriction is always present (or dilation due to anoxia from a severe overdose). Other neurological signs commonly observed are drowsiness, slurred speech, and impairment in attention and memory.

The maladaptive behavioral effects may include impaired judgment, interference with social or occupational functioning, and failure to meet responsibilities.

Associated features. Pupillary constriction may lead to reduced visual acuity. The inhibitory effect of the substance on gastrointestinal motility may cause constipation. There may be analgesia.

For many individuals, the effect of taking an opioid for the first time is dysphoric rather than euphoric, and nausea and vomiting may result.

Course. The factors involved in rapidity of onset and the duration of intoxication are discussed in the text on Intoxication as an Organic Brain Syndrome (p. 121).

A single dose of morphine taken intravenously will reach its peak subjective effect in 5 minutes or less. The effect lasts from 4 to 6 hours and is usually followed by a "down" feeling.

Complications. The most serious complication is opioid poisoning, manifested by coma, shock, pinpoint pupils, and depressed respiration, with the possibility of death from respiratory arrest. This syndrome can be rapidly reversed by intravenous administration of a narcotic antagonist such as naloxone, nalorphine, or levallorphan if this treatment is given before irreversible brain anoxia has occurred.

Other complications are similar to those of Alcohol Intoxication, but are less commonly seen (see p. 130).

Differential diagnosis. Other Substance-induced Intoxications may cause a similar clinical picture. Barbiturates and alcohol are most likely to be confused in this regard. **Barbiturate** and **Alcohol Intoxication** can be distinguished by the absence of pupillary constriction in the latter and by the use of laboratory tests. A mild **Hallucinogen Hallucinoses** may occasionally cause a similar picture. **Cocaine, amphetamines, and hallucinogens** cause pupillary dilation; but unless the individual is in a state of severe intoxication, the possibility that this dilation is a sign of Opioid Intoxication is unlikely.

Diagnostic criteria for Opioid Intoxication

A. Recent use of an opioid.

B. Pupillary constriction (or pupillary dilation due to anoxia from severe overdose).

C. At least one of the following psychological signs:

- (1) euphoria
- (2) dysphoria
- (3) apathy
- (4) psychomotor retardation

D. At least one of the following neurological signs:

- (1) drowsiness
- (2) slurred speech
- (3) impairment in attention or memory

E. Maladaptive behavioral effects, e.g., impaired judgment, interference with social or occupational functioning.

F. Not due to any other physical or mental disorder.

292.00 Opioid Withdrawal

The essential feature is a characteristic withdrawal syndrome due to recent cessation of or reduction in use of an opioid. The syndrome includes lacrimation, rhinorrhea, pupillary dilation, piloerection, sweating, diarrhea, yawning, mild hypertension, tachycardia, fever, and insomnia.

The symptoms and signs of Opioid Withdrawal may be precipitated by the abrupt cessation of opioid administration after a one- or two-week period of continuous use or by administration of a narcotic antagonist (e.g., naloxone or nalorphine) after therapeutic doses of an opioid given four times a day for as short a period as three or four days.

Associated features. Common associated features include restlessness, irritability, depression, tremor, weakness, nausea, vomiting, and muscle and joint pains. These symptoms together with the symptoms noted above resemble the clinical picture of influenza.

Depending on the observer and the environment, there may be complaints, pleas, demands, and manipulations all directed toward the goal of obtaining more opioids. A need for analgesia may be simulated or the withdrawal symptoms may be exaggerated.

Course. Mild Opioid Withdrawal may occur after abrupt withdrawal of analgesic doses of morphine after 7 to 10 days of administration. However, it is relatively uncommon for opioid craving to occur in the context of analgesic administration for pain from physical disorders or associated with surgery. The withdrawal symptoms with full-blown craving for opioids usually occur secondary to abrupt withdrawal in individuals dependent on opioids. In the case of morphine or heroin, the first withdrawal signs are usually noted within 6 to 8 hours of the previous dose, reach a peak on the second or third day, and disappear in 7 to 10 days.

The withdrawal syndrome from meperidine begins more quickly after the last dose, reaches a peak within 8 to 12 hours, and is over within 4 or 5 days. Methadone withdrawal may not begin for 1 to 3 days after the last dose, and then may be relatively mild. The symptoms are usually over by the 10th to 14th day.

Withdrawal symptoms from semisynthetic and synthetic opioids are qualitatively similar to those from morphine, the general rule being that substances with a short duration of action tend to produce short, intense, withdrawal syndromes whereas substances that are slowly eliminated produce withdrawal syndromes that are prolonged but milder. However, narcotic antagonist-precipitated

withdrawal following administration of long-acting substances can be quite severe.

Complications. Death rarely occurs unless the individual has a severe physical disorder, such as cardiac disease.

Differential diagnosis. **Influenza** is remarkably similar in its clinical picture to Opioid Withdrawal.

Other substance withdrawals and mixed withdrawals, especially **Barbiturate or Similarly Acting Sedative or Hypnotic Withdrawal**, can be differentiated from Opioid Withdrawal by testing of blood and urine and by taking a careful history, bearing in mind that individuals giving such a history are often unreliable. Third-party confirmation is extremely valuable.

Diagnostic criteria for Opioid Withdrawal

A. Prolonged, heavy use of an opioid (or administration of a narcotic antagonist following a briefer period of use).

B. At least four of the following symptoms due to the recent cessation of or reduction in opioid use:

- (1) lacrimation
- (2) rhinorrhea
- (3) pupillary dilation
- (4) piloerection
- (5) sweating
- (6) diarrhea
- (7) yawning
- (8) mild hypertension
- (9) tachycardia
- (10) fever
- (11) insomnia

C. Not due to any other physical or mental disorder.

COCAINE ORGANIC MENTAL DISORDER

Cocaine is usually applied to the mucous membrane of the nose by sniffing the crystalline flakes or powder. Intravenous administration is sometimes preferred, most commonly by opioid users who mix cocaine with heroin, a combination referred to as "speedball." Cocaine "base" is smoked in pipes or cigarettes, and has effects similar to cocaine taken intravenously.

There is apparently no withdrawal syndrome.

305.60 Cocaine Intoxication

The essential features are specific psychological and physical symptoms and maladaptive behavioral effects due to the recent use of cocaine. The psychological

symptoms typically include a sense of well-being and confidence, with heightened awareness of sensory input. There may be psychomotor agitation, elation, grandiosity, loquacity, pacing about, and pressured speech. The physical symptoms include tachycardia, pupillary dilation, elevated blood pressure, perspiration or chills, nausea and vomiting, anorexia, and insomnia. The psychological and physical symptoms begin no longer than one hour after administration, and may occur within a few minutes.

The maladaptive behavioral effects may include fighting, impaired judgment, and interference with social or occupational functioning.

Associated features. If taken intravenously or smoked, cocaine produces a characteristic "rush" of well-being and confidence. If the intoxication is severe, there is likely to be confusion, rambling or even incoherent speech, anxiety, and apprehension. There may be headache and palpitations. Intoxication with intravenous administration of high doses of cocaine may be associated with transient ideas of reference, paranoid ideation, a subjective sense of "profound thoughts," increased sexual interest, ringing in the ears, hearing one's name called, and a sensation of insects crawling up the skin (formication) or seeing insects. There sometimes are increased curiosity and bizarre behavior, such as sorting objects into various piles. Stereotyped movements of mouth and tongue are sometimes noted.

One hour or longer after the characteristic behavioral and physical effects have subsided, anxiety, tremulousness, irritability, and feelings of fatigue and depression often ensue. During this period, referred to as "crashing," there often is a craving for more cocaine.

Course. The course of Cocaine Intoxication is usually self-limited, with full recovery within 24 hours.

Complications. Syncope or chest pain may occur. There may be seizures following large doses. Death may result from cardiac arrhythmias or respiratory paralysis.

Differential diagnosis. **Manic episode** may present with symptoms similar to those seen in Cocaine Intoxication. **Amphetamine Intoxication** and **Phencyclidine (PCP) Intoxication** may cause a similar clinical picture, and can be distinguished from Cocaine Intoxication only by the presence of cocaine metabolites in a urine specimen or cocaine in plasma.

Diagnostic criteria for Cocaine Intoxication

A. Recent use of cocaine.

B. At least two of the following psychological symptoms within one hour of using cocaine:

- (1) psychomotor agitation
- (2) elation
- (3) grandiosity
- (4) loquacity
- (5) hypervigilance

C. At least two of the following symptoms within one hour of using cocaine:

- (1) tachycardia
- (2) pupillary dilation
- (3) elevated blood pressure
- (4) perspiration or chills
- (5) nausea and vomiting

D. Maladaptive behavioral effects, e.g., fighting, impaired judgment, interference with social or occupational functioning.

E. Not due to any other physical or mental disorder.

AMPHETAMINE OR SIMILARLY ACTING SYMPATHOMIMETIC ORGANIC MENTAL DISORDERS

This group includes all of the substances of the substituted phenylethylamine structure, such as amphetamine, dextroamphetamine, and methamphetamine ("speed"), and those with structures differing from the substituted phenylethylamine that have amphetamine-like action, such as methylphenidate or some substances used as appetite suppressants ("diet pills"). These substances are taken orally or intravenously.

305.70 Amphetamine or Similarly Acting Sympathomimetic Intoxication

The essential features are specific psychological and physical symptoms and maladaptive behavioral effects due to the recent use of amphetamine or similarly acting sympathomimetic substances.

All of the features (essential, associated, course, complications, and differential diagnosis) are virtually identical with those of Cocaine Intoxication. One exception is that delusions or hallucinations are always transient in Cocaine Intoxication, whereas in the intoxication due to amphetamine or similarly acting sympathomimetic substances, they may persist beyond the time of direct substance effect. When this occurs, the syndrome is then referred to as an Amphetamine or Similarly Acting Sympathomimetic Delusional Disorder (p. 149). The other exception is that on occasion an intoxication from this class of substances may develop into a Delirium.